Skills Summary

Subtraction and Negative Signs: Diagnosing the Error

Use this reference while completing your worksheet or when studying.

Skill 1 — Subtracting Signed Numbers

Rule — add the opposite: $p - q = p + (-q)$

- $p - (-q) = p + q$ (subtracting a negative moves *right*)
- $p - q = p + (-q)$ (subtracting a positive moves *left*)
- Same signs after rewriting: add the distances, keep that sign.
- Different signs after rewriting: subtract the distances, keep the sign of the farther-from-zero number.

Example: Evaluate $-6 - (-2)$.
Step 1. The second number is negative, so "add the opposite" turns the subtraction into an addition of a positive.
Step 2. $-6 - (-2) = -6 + 2$; the signs differ, so $6 - 2 = 4$ with the sign of -6, giving $\mathbf{-4}$.

Try it: Evaluate $-9 - (-4)$.

check: -5

Skill 2 — Compare Two Signed Expressions

Compare by rewriting, not by computing. Turn each side into "first number + signed number," then read the two forms.

Two expressions are equal exactly when they add the same signed number to the same start. On the number line, *left is less*: $-9 < -5 < 5$, even though $9 > 5$ as digits.

Example: Put $<$, $>$, or $=$ between $-5 - (-2)$ and $-5 + 2$.
Step 1. Both sides start at -5, so rewriting the left side as an addition will make the comparison readable without arithmetic.
Step 2. $-5 - (-2) = -5 + 2$, which is the right side character for character, so the relation is $=$.

Try it: Put $<$, $>$, or $=$ between $3 - 8$ and $3 + 8$.

check: $<$

Skill 3 — Find and Fix a Sign Error

Diagnose from the answer, in this order:

- Answer too far *left*? The double negative was dropped: $-(-q)$ was read as $-q$.
- Answer too far *right*? A subtraction of a positive was turned into an addition.
- Answer correct in size but positive? The two numbers were reversed to avoid a negative.

Size check first: subtracting a negative always increases the value; subtracting a positive always decreases it.

Example: A student writes $-1.5 - 3.5 = 2$. Find and fix.
Step 1. Subtracting a positive must move left, so an answer larger than -1.5 fails the size check — the sign of the second term was flipped.
Step 2. Correctly, $-1.5 - 3.5 = -1.5 + (-3.5)$; same signs, so the distances add to give $\mathbf{-5}$.

Try it: A student writes $-2.75 - (-5) = -7.75$. Find and fix.

check: **2.25**